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Data Analytics, Innovation, and Firm Productivity

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Abstract. We examine the relationship between data analytics capabilities and innovation using detailed firm-level data. To measure innovation, we first utilize a survey to capture two types of firm practices, process improvement and new technology development for 331 firms. We then use patent data to further analyze new technology development for a broader sample of more than 2,000 publicly traded firms. We find that data analytics capabilities are more likely to be present and are more valuable in firms that are oriented around process improvement and that create new technologies by combining a diverse set of existing technologies than they are in firms that are focused on generating entirely new technologies. These results are consistent with the theory that data analytics are complementary to certain types of innovation because they enable firms to expand the search space of existing knowledge to combine into new technologies, as well as the theoretical arguments that data analytics support incremental process improvements. Data analytics appears less effective for developing entirely new technologies or creating combinations involving a few areas of knowledge, innovative approaches where there is either limited data or limited value in integrating diverse knowledge. Overall, our results suggest that firms that have historically focused on specific types of innovation—process innovation and innovation by diverse recombination—may receive the most benefits from using data analytics.

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Introduction

As the availability of data about consumers, suppliers, competitors, and partners proliferates, firms are expanding the use of large-scale analytics to make decisions (Brynjolfsson et al. 2011). Part of this trend involves the exploitation of operational data that is provided as a by-product of the deployment of enterprise information systems (McAfee 2002, Aral et al. 2012). New types of data are also proliferating such as individuals' activities, locations, relationships, clickstreams, and keyword searches, all of which can offer comprehensive information about consumer behavior in real time, eliminating the need to conduct one-off surveys with small samples of consumers (Wu and Brynjolfsson 2014). These data are even more valuable when combined, affording an unprecedented opportunity to influence business decisions.

Initial research into the value of data-driven decision making suggests that analytics can have considerable effects on firm performance (McElheran and Brynjolfsson 2015, Müller et al. 2018). One area that has begun to attract renewed interest in the IT value literature is innovation and how data analytics can be used to facilitate innovation. While process-related innovation has long been tied to IT investment, such as in theories and practices related to "reengineering" (Hammer 1990), only a few studies have documented that IT may also support the innovation process more generally (Gao and Hitt 2012, Kleis et al. 2012). There may be an especially close tie between analytics and innovation beyond just process improvement because the acquisition of data can facilitate the creation of knowledge (Joshi et al. 2010, Majchrzak and Malhotra 2016). However, as with prior generations of IT, it is likely that analytics supports some types of innovation better than others. For instance, a significant benefit of enterprise software systems and social media technologies was the acquisition and distribution of data about various business processes that enabled data-driven process improvement (McAfee 2002, Hitt et al. 2015).

It is plausible but unknown whether data analytics, as it is currently deployed, has the same effects on other types of innovation processes such as new technology development. Some technologies are created by combining a diverse set of previous technologies in new ways (we will refer to this as "diverse recombination"). There is some evidence that analytics can support the search process of identifying relevant technologies and exploring new combinations—this would suggest a relationship between analytics and recombination approaches that integrate a diverse collection of prior technologies. Other technologies are completely novel, differing substantially from prior art (we will refer to this technology development process as "novel" innovation). It is less known whether analytics can be useful for the development of novel innovation. Indeed, there may even be evidence to the contrary: Steve Jobs notably refused to do any customer surveys prior to the introduction of many new Apple products such as the iPad, and yet the sales of media tablets in general vastly outperformed analyst expectations.

Innovation strategy research has long recognized the distinction between the creation of new knowledge and the reuse or reapplication of existing knowledge, and has shown that the types of organizational processes that support different types of technology development are fundamentally different (March 1991, Benner and Tushman 2003). As a result, it may be difficult for firms to efficiently reuse knowledge and create completely new knowledge at the same time. Understanding the effect of data analytics on different types of innovation and how organizational practices may moderate these relationships are especially important in light of recent data that suggest an "innovation slowdown" despite the increase in technologies that support knowledge workers such as data analytics (Bloom et al. 2017).

We hypothesize that data analytics can help innovation that requires intensive information processing and search such as process improvement and recombination from diverse sources of prior technology. More specifically, process improvement and diverse recombination are complementary to analytics—firms that engage in innovation through process improvement and diverse recombination will show a greater demand for data analytics and have a higher return from their analytics investments. By contrast, we expect data analytics to have smaller effect (if any) on the technology development for novel innovation that may not require or benefit from broad search or where there are no data on which to apply analytics capabilities.

Our identification approach relies on the observation that there has been a rapid and exogeneous shift in the amount of data available to firms and in the capability of tools to process these data (Tambe 2014), while innovation strategy is likely to change much more slowly. As different firms alter their data analytics capability in response to changes in the amount of data and their ability to hire and absorb personnel with analytics skills, there will be cross-sectional and time series variation in the matching of data analytics skills to innovative practices that enables the identification of complementarities relationships. To address potential endogeneity in analytics skills relative to firm performance, we develop instruments based on the availability of analytics skills in closely related firms in the hiring network (Hitt et al. 2015, Wu et al. 2018) and address the general endogeneity of production inputs using panel data techniques from the microproductivity literature.

We need to measure two things: the capabilities of firms in using data analytics skills and types of innovations in firms. We measure data analytics capabilities of a firm as the proportion of employees who possess data analytics skills. To do this, we use a large-scale résumé database covering 6 million résumés from 1988 to 2007 and an online job review platform covering 3.7 million job reviews from 2008 to 2013. We use two data sources for measuring innovation. First, we use survey data on 331 large firms conducted in 2008 to obtain information on internal innovative processes that includes both process improvement and general technology development. Second, we use patent data to measure technology development practices for more than 2,000 publicly traded firms from 1988 to 2013. Using well-developed models and metrics utilized in the R&D literature, we can distinguish between different types of technology development in patents (identifying firms that engage in diverse recombination or novel innovation). Measurements using patents can also broaden the sample and enable the use of panel data methods, which is especially important for examining processes that evolve over time and for addressing unobserved heterogeneity in firms.

We then combine our innovation and analytics data with production data from Compustat similar to that used for prior research in IT and R&D productivity (covering the period 1988–2013) to enable estimation of complementarities (Milgrom and Roberts 1990, Athey and Stern 1998): (1) demand models that show the conditional correlation between analytics and innovative practices, and (2) productivity models that examine whether certain combinations of practices are associated with increased performance.

Overall, we find that firms engaged in activities that require intensive information processing and search—incremental process innovation and patent activity indicating diverse recombination—have a higher demand for data analytics skills, and the combination of analytics with these practices is associated with higher productivity. However, firms engaging in developing completely new technologies (a new concept or technology class) show no effect from using data analytics. These effects are robust to several different econometric and measurement approaches and are above any contribution of innovation or analytics investments more generally or from IT labor.

Theory and Literature

The Business Value of Analytics

Data-driven decision making has become more widespread as data availability has grown. The work of Brynjolfsson et al. (2011) is one of the first large-scale studies on the impact of using data to make decisions, and they found considerable returns generated by the adoption of data-driven decision-making practices. Similarly, Saunders and Tambe (2013) found that firms more focused on data use at an executive level—as measured by discussion of data-related topics in their annual reports—have higher productivity and market valuations. Analysis of manufacturing firms using census data also shows that the rapid adoption of data-driven decision making is associated with productivity gains (Brynjolfsson and McElheran 2016).

Since the mid-1990s, the IT literature has recognized that IT is a significant driver of productivity only when firms have also adopted certain complementary organizational practices (Hitt and Brynjolfsson 1996, Bresnahan et al. 2002, Melville et al. 2004, Bapna et al. 2013). The primary explanation for these complementarities is that they affect the firm's ability to gather and act on information for internal operational decisions; more recent research has extended these arguments to information external to the firm (Tambe et al. 2012) including information gleaned from social media (Hitt et al. 2015). Finally, the practice of data-driven decision making is shown to complement to other investments in IT and the use of team-based decision making (McElheran and Brynjolfsson 2015).

Not all decisions are amenable to data-driven decision making. From the early days of the IT value literature, it has been recognized that different types of information and the information processing capabilities can be either substitutes for or complements to managerial judgment and employee skills (Leavitt and Whisler 1958, Zuboff 1988, Hammer 1990). For instance, skill-biased technical change research shows that the introduction of information systems and related organizational practices can affect occupations differently (Levy and Murnane 1996, Murnane et al. 1999, Autor 2015). In some cases, routinized decisions, which may or may not be done by knowledge workers, can be replaced entirely by information systems, but, in other cases, aggregation of data can enable more tasks to be performed by a single person, broadening decision authority (Hammer 1990). These trends have received renewed interest as we are observing the effects of the "digitization of work" across increasingly wide domains (Brynjolfsson and McAfee 2014). While the use of data analytics has nonetheless expanded into new domains, there are still areas that are not amenable to automation, such as tasks involving creativity or insight (March and Simon 1958, McAfee and Brynjolfsson 2017). Regardless, at any given time and state of technology, certain types of decisions and activities, including innovation, are better supported by data and analytics tools than others.

The innovation literature also documents extensive evidence that different organizational practices support different methods of knowledge creation (March 1991, Benner and Tushman 2003, Kaplan and Vakili 2015), suggesting that the organizational practices that support new knowledge creation are different from (and perhaps in conflict with) the organizational practices that promote reuse of existing knowledge. The IT-innovation literature has made a similar distinction between incremental and radical innovation and provides a parallel set of arguments that suggest a limited ability to successfully pursue different types of IT-enabled innovation (Pavlou and El Sawy 2006, Lee et al. 2015).

As the use of data analytics is growing, the benefits of analytics will likely vary considerably across firms. To our knowledge, the prior literature has neither examined the relationship between innovation and analytics investments nor identified the types of innovation that might be complementary to analytics. It is thus important to understand how to effectively use analytics and how some firms may reap greater reward from using analytics to innovate.

Data Analytics, Search, and Innovation

Process improvement has a long history of being closely tied to IT usage, such as the reengineering wave that followed the deployment of enterprise-wide IT (see, e.g., Hammer 1990). The use of data in managing and refining processes has emerged as an important competitive advantage because it can help firms become more flexible and responsive to the business environment and support reductions in production cost, risk, and process variance (Frei et al. 1999, Mithas et al. 2011, Barua et al. 2012).

The types of processes amenable to data-driven decisions have expanded. For instance, human resource managers are increasingly reliant on data and HR analytics for matching both internal employees and external candidates to the right jobs (Bock 2015). Data availability and accuracy can be critical to ensuring forecasting and streamlining supply chain management processes for retailers (Fisher et al. 2000). Data management and information processing capabilities are important for business process improvements and interfirm coordination necessary for effective implementation of outsourcing arrangements (Tanriverdi et al. 2007, Mani et al. 2010). Traditional industries such as agriculture have also been transformed by geolocation services that improve the precision of planting and application of agrichemicals, as well as leveraging real-time monitoring systems that improve yield and quality forecasts (Mucherino et al. 2009). As data availability grows through the digitization of various human and economic behaviors, firms across various industries can benefit substantially from using these data. Real-time analytics can be used for fine-tuning existing processes as well as identifying any process limitations that hinder the successful execution of firm strategy. Thus, we hypothesize that the use of data analytics can improve business processes, and that the joint adoption of both analytics and process-oriented firm practices can improve productivity.

Hypothesis 1 (H1). *Data analytics and process improvement-related practices are complements.*

IT should lead to greater innovative output more broadly (Sambamurthy et al. 2003, Pavlou and El Sawy 2006, Joshi et al. 2010, Bardhan et al. 2013, Chakravarty et al. 2013) and is shown specifically to increase the rate of patent output and trademark activity (Gao and Hitt 2012, Kleis et al. 2012). However, it is likely that analytics provides greater support for some types of innovative activity than others because firms have different innovation preferences. Perhaps the clearest link between analytics and innovation is the increase in the ability to expand the search space for information and to combine existing knowledge to create new knowledge, especially when considering knowledge in previously distant and unconnected domains. We refer to this innovation produced in this manner as *diverse recombination* (or *diversity*), indicating the variety of knowledge brought together to create a new innovation. This recombination process has been shown to have important implications for generating high-impact innovations (Hargadon and Sutton 1997, Audia and Goncalo 2007). While it is often difficult to link distant domains together, analytics capabilities automate the combination of diverse ideas to some extent and manage the associated complexity by tracking what has been explored and what is already known. For instance, IBM's Watson computer needed only 1 month to digest 23 million medical papers from many different disciplines to identify 6 new proteins that are linked to many types of cancer suppressors, whereas it has taken 30 years for human researchers to identify only 28 such proteins (Chen et al. 2016). Similarly, chemical firms such as BASF are applying artificial intelligence to analyze scientific papers and other sources of materials information to design new materials, reducing the cost and uncertainty of advanced materials development (Mullin 2017). McAfee and Brynjolfsson (2017) provide a variety of examples of technology-enabled innovation. Some of these represent fundamentally new ideas, but most involve refinement or new combinations of existing ideas rather than the creation of entirely new technologies or products. Thus, by expanding the search space for possible technologies and accelerating the rate at which this information can be processed, data analytics can substantially improve the ability to combine diverse sets of knowledge together. This leads to the following hypothesis:

Hypothesis 2 (H2). *Data analytics and diverse recombination are complements.*

In contrast to innovation that involves the reuse or recombination of existing knowledge, some innovations have no precedent, creating entirely new classes of technologies or products. These types of innovations may be valuable on their own and also generate increasing returns to other forms of innovation as they expand the space of recombination possibilities by adding new technology building blocks (Akcigit et al. 2013).[1] We will refer to this technology development process as *novel* innovation because it involves the generation of a new technological trajectory rather than the extension of existing ones (Kaplan and Vakili 2015). As noted earlier, the link between data analytics and novel innovation is likely to be weaker than the link between analytics and diverse recombination or analytics and process improvement. On the one hand, improvements in the innovative process may increase the ability to create novel innovations as much as any other type of innovation. Some firms, such as Capital One, have created new products by conducting large-scale customer experiments with a wide variety of new products and then employing analytics to identify the small subset of product designs that may be profitable (Clemons et al. 1995). On the other hand, this type of mass new product experimentation is difficult for most firms, which limits the efficacy of this type of strategy. In addition, prior work suggests that the creation of novel innovation often involves deep understanding of a narrow domain rather than a broad but shallower search across many domains—the organizational structures, resources, and incentives for these different approaches may be different or even in conflict. For instance, the ability to easily find incremental improvements can compete with the incentives for identifying completely new ideas (Weisberg 1999, Taylor and Greve 2006).

There may also be inherent limits to the current state of analytics technology for supporting novel innovation. The limitations of computers for certain types of cognitive tasks such as those requiring creativity or insight have long been recognized (see, e.g., Simon 1966), and they may not be solvable with the types of data that are increasingly available. Analyzing data captured through various businesses processes and consumer behaviors at the finest-grain level may lead to evolutionary rather than revolutionary innovations (Moon 2010) because the generation of novel innovations relies much more on tacit knowledge and other types of information that are less amenable to quantitative analysis (Henderson and Clark 1990, Von Hippel 1994, Nonaka and Von Krogh 2009). It could also be that certain types of data (e.g., expressed preferences in surveys) may have limited value for novel innovation because they may not reflect true attitudes or behaviors (Avery and Norton 2014), especially for new products not previously encountered.

Improved information also may not lead to novel innovation because of biases in managerial decision making. For instance, mining consumer observations for common patterns, regardless of how detailed those observations may be, often serves the purpose of confirming managers' preexisting beliefs about consumers rather than exploring new hypotheses (Martin 2009). Experimental methods may help close this gap, but these experiments are costly, except for in firms in internet retail or online services, where A/B testing has become commonplace (Clay et al. 2001). Regardless, such methods still require the generation of novel ideas to test. Thus, while data analytics can support broader search across existing knowledge and improve the ability to handle large-scale data (such as those generated by customer interactions), these capabilities may not be directly applicable to the process of creating novel innovations where there are fewer precedent technologies and data. However, we are not taking a strong position that data analytics would hurt novel innovation, but hypothesize that analytics helps less with novel innovation than innovation that relies on search and intensive information processing:

Hypothesis 3 (H3). *Data analytics is less likely to complement the creation of novel innovation than the creation of innovations that are reliant on search or information processing.*

Data and Measurement

Process and Innovation Practices

To gain some preliminary insight regarding the relationships between analytics and different innovative practices, we utilize a survey administered to senior human resource managers and chief information officers from large publicly traded firms in 2008. The survey is a joint effort between two academic institutions and a major global consulting firm, and was focused on identifying the relationship between business technology usage (including the use of analytics), firm organization, and a variety of internal and external firm processes (including innovation). In total, 331 firms responded to the survey. The survey asks about various business and IT practices, drawing questions in part from two previous surveys on IT usage and workplace organization practices conducted in 1995 and 2001 (Brynjolfsson and Hitt 1996, Brynjolfsson et al. 2011). The survey added questions about innovative and process improvement business practices, as well as general decision-making practices. This survey is especially useful for understanding business process innovation that is difficult to observe in other innovation-related data such as patent activity. To explore what types of organizational practices data analytics can complement, we use survey responses to construct two organizational practices scales as shown in Table 1. Most of the business practice questions were captured on a 5- or 7-point Likert scale (for the 7-point scale, a typical question had a mean on the order of 3–4 and a standard deviation of approximately 1; see Table 1). There are also a few questions where companies list their strategic priorities, indicating whether they are related to process or innovation. The responses to these questions are converted to binary variables.

We measure practices related to process improvement as a composite scale of 7 process-related questions. Four questions focus on the extent to which refining and improving existing processes is important to the firm (P1–P4). The other three questions (P5–P7) ask about the extent to which a firm monitors, tracks, and analyzes processes. This scale, as well as those that follow, were constructed as the standardized sum of the standardized values of the scale components (we replace missing data on individual responses with the sample mean of the other firms' response [Santos 1999]). Correlations between the 7 constructs within this scale are all positive, and the Cronbach's alpha is 0.74. Thus, we calculate

$$Process\ Practices = Norm(Norm(P1) + Norm(P2) + \ldots + Norm(P7)). \quad (1)$$

To gauge a firm's general innovation practices, we use 7 additional survey questions. The first 2 questions (I1 and I2) measure how fast the focal firm can introduce an innovative product or service, and the next 3 questions (I3–I5) measure how important innovation is to the firm. The last 2 questions (I6 and I7) measure the likelihood that a firm is a leading-edge adopter of technology. Being a successful lead user of new technologies can indicate innovation capabilities because the integration of new technologies often requires the conception and introduction of new business processes.[2] The same approach is used to construct the scale as the process improvement measure. The Cronbach's alpha of this scale is 0.48. The relatively low value reflects the multidimensionality of

Table 1. Process and Innovation Practices

Survey question	Observations (firms)	Mean	Standard deviation	Minimum	Maximum
Process practices					
P1 To what extent does the following statement describe the work practices and environment of your entire company?: We have a strong ability to **make incremental changes or improvements to our business processes**.	4,584 (206)	3.594	0.881	1	5
P2 Please list the most important core activities for your primary business: **Process development, process quality, process management and improvement**	4,358 (196)	0.054	0.226	0	1
P3 Currently, how effective is your IT organization in proactively engaging with business leaders to **refine existing processes and systems**?	4,535 (204)	3.044	0.984	1	5
P4 To what extent do the following statement describe the work practices and environment of your entire company?: We have a cross-functional process orientation.	4,549 (204)	3.151	1.023	1	5
P5 Analysis and improvement are part of everyone's job	4,530 (203)	3.561	1.027	1	5
P6 We rigorously track and ensure capture of the benefits or returns specified in business cases	4,537 (204)	2.866	0.971	1	5
P7 We monitor performance of activities through a set of well-defined metrics	4,562 (205)	3.365	1.002	1	5
Process = Norm(*Norm*(*P*1) +...+ *Norm*(*P*7))	4,631 (208)	0.000	0.609	−1.879	1.559
Innovation practices					
I1 On average, how long does it take your organization to perform the following activities?: **Design, create and introduce a new product or service** after approval.	4,130 (185)	3.468	1.394	1	7
I2 On average, how long does it take your organization to perform the following activities?: **Introduce a new production technology or method.**	3,922 (176)	3.444	1.192	2	7
I3 List the most important core activity for your primary business unit: **product development, product design, innovation**	4,358 (196)	0.167	0.368	0	1
I4 To what extent do the following statement describe the work practices and environment of your entire company?: Our company **stresses innovation** over operational excellence (efficient execution).	4,571 (205)	2.691	1.061	1	5
I5 To what extent do the following statements describe the work practices and environment of your *entire company*?: We encourage experimentation to continually improve our offerings	4,510 (202)	3.052	1.029	1	5
I6 To what extent do the following statements describe the work practices and environment of your *entire company*?: We are usually the **leading-edge adopter** of new technologies in our industry.	4,518 (203)	2.624	1.183	1	5
I7 How would you generally characterize your company's approach to use technology in support of business capabilities? (Choose one): (a) Leading edge adopter; (b) Fast follower; (c) Mature adopter; (d) Late adopter	5,504 (247)	2.482	0.832	1	4
Innovation = Norm(*Norm*(*I*1) +...+ *Norm*(*I*7))	6,336 (329)	0.000	0.523	−1.894	1.294

innovative practices—firms adopting any one practice do not necessarily adopt all of the others—and reflects the fact that different types of innovation may be more important to firms in different industries or competitive environments. We separately estimate the subcomponents of the innovation practices variable, and they show consistent results, so it does not appear that our results are biased by using an aggregate scale (see Online Appendix C).

$$Innovation\ Practices = Norm(Norm(I1) + Norm(I2) + \ldots + Norm(I7)) \quad (2)$$

The survey has the advantage that it measures many firms' innovation practices directly, but has the disadvantage of focusing on a single snapshot in time for a limited number of firms and on innovation generally rather than specific types of innovation. To explore in greater detail how data analytics affects innovations in firms, we explore patent characteristics of more than 2,000 publicly traded firms that were issued a patent between 1988 through 2013. This covers a much broader range of companies and industries than the 331 firms surveyed, can be used to classify different types of innovation more precisely, and can capture the changes over time. Furthermore, both surveys and patents can be used to independently support our central hypotheses about how data analytics affects innovation. The distribution of process- and innovation-oriented practices is shown in Figure 1.

Patents

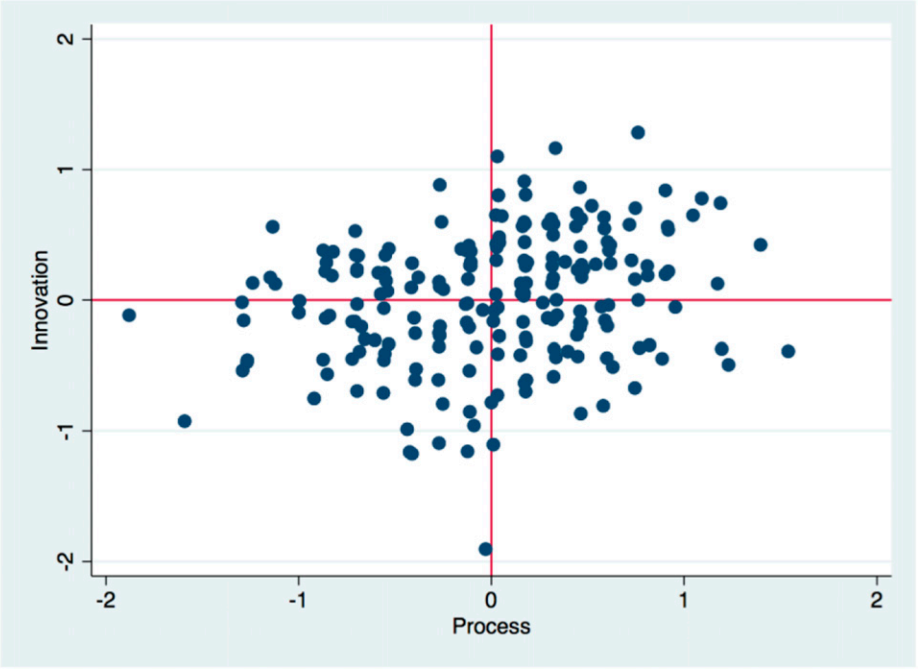
To identify diverse recombination and novel innovation, we construct measures based on each firm's patent characteristics. Patent data have been extensively used to understand innovation and the process of technical change (Griliches et al. 1986, Griliches 1990, Hall et al. 2001), and they provide a number of specific advantages for studying firm innovation. The patent examination process in the United States is an objective evaluation by a government entity. This fact may make measurement more consistent. The United States Patent and Trademark Office (USPTO) evaluates patents on specific criteria: novelty, usefulness, and non-obviousness (Hall et al. 2001). These standards have been stable over time, making comparison across firms and years highly robust. In addition to providing a description of the technology, patents have detailed records of citations that show the relationship to prior technologies, identifiers indicating the general area of technology (patent class) described in the patent.

We primarily use the NBER Patent Citation Data File (Hall et al. 2001), which links patents to publicly traded firms through 2006.[3] To match patents from 2007 to 2013, we obtain the original patent data from the USPTO and use the same methods as Hall et al. (2001) to match patents to firms. These methods account for potential problems in matching such as name misspellings or patents assigned to subsidiaries. Following the convention in the R&D literature (Griliches et al. 1986, Hall et al. 2005), we use the application year (as opposed to the issue year) because it more closely approximates the time at which the firm produced and had available the innovation described in the patent. To construct a total stock of patents, we capture all patents issued to the firm in the 20 years prior to the observation year, and we form a stock measure by tallying the number of patents and then assuming an annual 15% depreciation, a value suggested in the R&D productivity literature (Hall et al. 2001, Hall et al. 2005).

Diverse Recombination ("Diversity")

To measure the extent to which a patent represents a recombination of a diverse set of existing knowledge, we begin with a measure created in Hall et al. (2001) [also described as "technological diversity" (Kaplan and Vakili 2015)] that is a Herfindahl concentration

Figure 1. (Color online) Distribution of Firms on the Process and Innovation Practices



Notes. Each point in this scatter plot corresponds to a firm. All data points are distributed in the four quadrants and along the process (x-axis) and innovation (y-axis) practices. Table 1 shows more details about how process and innovation practices are measured.

index, calculated based on the distribution of different 3-digit main classes of prior art that are cited by the focal patent. To account for potential biases in this measure for patents with a small number of prior art references (that is, firms with fewer patents necessarily cite fewer categories of prior art), we adjust the index by the number of cites (Hall et al. 2001, pp. 44–46):

$$Diversity = \left(\frac{Cites}{Cites - 1}\right)\left(1 - \sum_{j}^{n_i} s_{i,j}^2\right), \tag{3}$$

where $s_{i,j}$ represents the percentage of citations that patent i cites that belong to patent class j, out of n_i different patent classes. Patents that cite a wider variety of classes will have a measure closer to 1 indicating higher diversity (the scale is bounded below by 0). Thus, the diverse recombination metric represents the variety of prior art that led to the focal innovation. We use a backward-citation-weighted average of firm patent stock to measure the diversity score for the entire firm using a perpetual inventory method approach with a 15% annual depreciation rate (see further detail in Online Appendix A).[4] Because our models control for the overall number of patents, the marginal effects of this measure can be interpreted as the effect of diversity rather than a mix of diversity and total innovative output, as would be suggested by the construction.

Novel Innovation ("Novelty")

To measure novelty, we note that patents contain a considerable amount of free-form text data describing the innovation, which can be used to represent the underlying inventive ideas. Prior research on patent innovation, including in information system research, used bibliometric methods and citation-based analysis to study innovation impact and evolution (Hall et al. 2001, Saunders 2011, Kleis et al. 2012). However, the raw text from the patent has been less frequently used, although a few studies have considered textual analysis such as counts, factor analysis, keyword co-occurrence, and topic modeling to identify the nature of the innovation described by the patent (Azoulay et al. 2007, Upham et al. 2010, Wu 2013, Kaplan and Vakili 2015).

Our measure is based on the idea that novel ideas can often be detected through vocabulary shifts (Kuhn 2012, Kaplan and Vakili 2015). In our case, we examine the appearance of new words or topics in the patent abstract. Specifically, we apply a "bag of words" model to identify patent terms and then calculate the age of each word by when it first appeared in the prior art in similar technological domains (Hasan et al. 2009, Boudreau et al. 2016). Each (nonstop) word in the patent is assigned an age. A word that has not appeared previously is assigned an age of 0. If the word has appeared in an earlier patent, it is assigned an age that is the difference between the application date of the focal patent and the date of earliest patent containing that word. The overall novelty score is therefore the average age of the words in a patent. We add 1 to the word age to avoid the possibilities of division by 0.

$$Novelty = \frac{1}{N}\sum_{w=1}^{N}\frac{1}{Age_w + 1} \tag{4}$$

The novelty of a firm's patent portfolio for a particular year is equal to the weighted average of the novelty score of the firm's own patents, accumulated over time and depreciated at an annual rate of 15%.

We prefer keyword-based measurement because it is likely to be more sensitive to shifts in vocabulary than are topic modeling methods such as latent Dirichlet allocation (LDA), which relies on the identification of a relatively stable set of topics for classification (Blei et al. 2003, Shaparenko and Joachims 2009). Because a stable ranking of topics is essential for accurate classification using LDA (Greene et al. 2014), the shifting vocabulary likely makes it difficult to find those rankings. In addition, the large topic space (essentially the language of all scientific endeavor over the span of 20 years) would make it difficult to establish a reasonable topic space without expert input, which defeats the benefits of automated classification. By contrast, the keyword-based analyses are relatively simple and have long been established to detect novelty at both the sentence and document levels (Voorhees 2003). They do not require auxiliary assumptions about the number of topics or degree of association between words needed to define a topic. As a check, we also run an LDA model to identify topics in patents and measure the novelty of each topic in patents according to when it was first detected. The topic modeling results are directionally consistent. However, we are more comfortable with the keyword-based approach because it imposes fewer assumptions.

We also create an additional alternative measure of novelty: whether an innovation involves the creation of a new technology class in the patent classification system. As new technology classes are often added to the patent classification, the patent that creates a new patent class can be viewed as a novel innovation that is radically different from the past. Our results using this novelty measure are similar to the vocabulary-based metrics (Online Appendix C).

Because the introduction of new vocabulary or the creation of new patent classes is relatively infrequent, these measures may be less sensitive to more subtle forms of novel innovation—in practical terms, this would reduce statistical power or generate greater errors in measurement. The use of a few different measures of novelty can help determining whether any absence of effects is a true effect or due to difficulty in measurement.

Data Analytics

We broadly define data analytics as the ability to process, analyze, and transform data to detect patterns, find useful insights, and support decision making. Analytics skills can encompass a variety of skills including data cleaning, data transformation using statistics and programing tools, and advanced skills in machine learning and artificial intelligence in general (see Online Appendix A). These wide-ranging skills are necessary for firms to take advantage of their digital data. To measure a firm's data analytics capabilities, we calculate the total number of employees with the relevant data analytics skills for each firm from more than six million individual résumés that provide the career histories of these individuals from 1988 to 2007. We also collect more recent data about data analytics skills from a large online job review website from 2008 to 2013 that contains job titles of current and former employees who provided employer reviews.

Our résumé data are similar to other large sample résumé datasets used in the IT value literature that have been shown to provide measures of information technology input that are similar to alternative data sources (such as the Harte-Hanks Computer Intelligence Technology Database) but provide reasonable sample coverage outside Fortune 1,000 companies (see, e.g., Tambe and Hitt 2013a for a more detailed discussion of the advantages and limitations of these datasets generally for IT research). Using natural language processing techniques, we identify the skills of each employee as reflected in the free-form text part of their résumé as well as the words that appear in their job titles. The text-mining algorithm maps free-form text responses to a taxonomy of skills (this includes direct matches on keywords like "analytics" as well as inferred matches on phrases such as "regression modeling"; see Online Appendix A for details). For each employee, we identify whether a data analytics–related skill is present in his or her résumé and then aggregate the total number of employees with such skills for each firm. We can approximate the timing of when the person acquired and actively began using the data analytics skills from the job history in the résumé. We then measure firm-level data analytics capabilities in the sample by aggregating all employees with data analytics for each firm in each year. To estimate the actual number of employees with data analytics skills from the sample of résumés, we assume that the observed employees are sampled from the underlying population of all workers. We also assume that firm and occupation-specific factors with respect to the likelihood of posting career trajectory or job reviews are uncorrelated.[5] Thus, our firm-level sample rate could be estimated by $\theta_j' = x_j/L_j$, where x_j is the number of employees in all occupations in our sample for firm j, and L_j is the total number of employees for firm j as provided by Compustat. Finally, data analytics skills at firm j are estimated by scaling the number of data analytics employees who are found in-sample at firm j by the sampling rate θ_j'.

To obtain data from 2008 to 2013, we use a large online job review platform containing 3.7 million job reviews from current and former employees that include information on job title, employer name, and length of employment. Unlike the résumé data, we do not have the free text job descriptions in the job review, but we can identify data analytics skills and IT skills through the job titles of each employee and aggregate them at the firm level. This generates a panel data of employee skills from 2008 to 2013. Specifically, we use the job title classification from O*Net to identify data-related positions, similar to how IT labor is distinguished from other employees in earlier work (Tambe and Hitt 2013b).[6] If any of the skills listed under the job title is related to data analytics, we assume that anyone with that job title has data analytics skills. We aggregate these individual-level skills for each firm-year observation, tracing back to the year when the employment spell started and using the length of employment indicated. We then similarly adjust the raw number of employees with data analytics skills by the underlying sampling rate, as we did with the résumé data. Thus, our data cover the critical period for the rising demand in data analytics for the 26 years from 1988 to 2013. We also complement our title-based classification by identifying a few relevant phrases about data analytics ("data," "statistic," and "business analytics") in job titles.

Lastly, we ensure that merging the two data sources produces comparable metrics by calculating the appropriate scales to convert measures from one data source to the other. Specifically, we use the overlapping period (2008–2010) between the résumés (we have a set of résumés collected in 2010 that have job titles but not detailed job descriptions) and the job review data to calculate a firm-specific scaling (Online Appendix A). We also separately estimate the models for each data source, and our results are consistent with the combined sample, suggesting that scaling and merging the metrics do not influence our results. In analyses of our pooled data set, we include a year dummy for each year of observation, which also implicitly controls for data source (since the samples do not overlap in time). Furthermore, a Chow test on our main results cannot reject the hypothesis that the data analytics coefficients are equal across our two data sources,[7] ensuring that we can comfortably combine our data into a single panel. To address the possibility of bias due to different methods of classifying skills in each data set, we also employ a single classification method using job titles in both the résumés and the job reviews; the results are similar.

Overall, the distribution of data analytics skills spans multiple job titles and functional areas. The yearly trend in data analytics is shown in Figure 2. We observe a gradual increase in data analytics skills after 1990 with a steeper ascent after 2004, and a small dip in 2013.

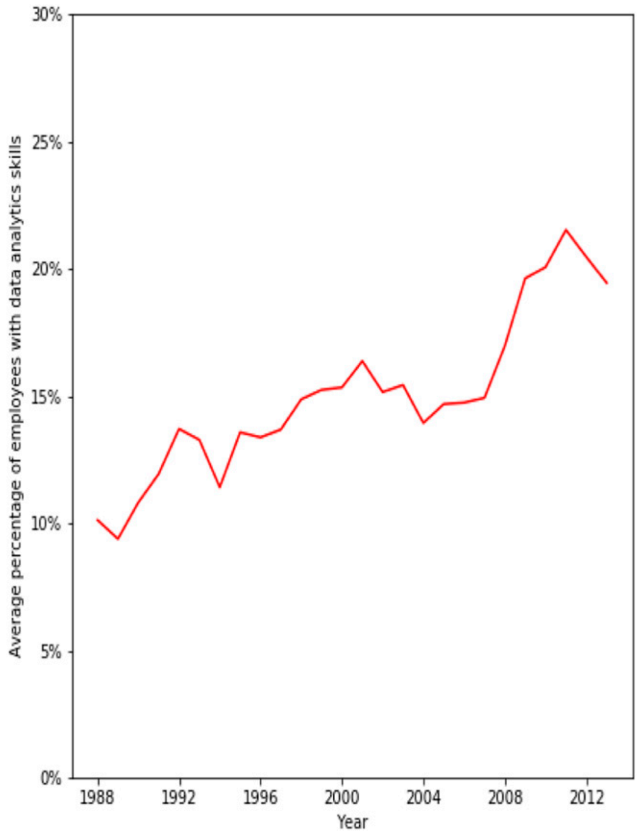
IT Labor

As a control, we also construct a measure of IT labor using a similar approach to Tambe and Hitt (2013a). IT labor is identified either by job titles that are clearly related to IT (for example, software engineer, systems analyst, programmer analyst) or that contain keywords that suggest the employee is an IT worker (for example, computer, website, software). We conduct various alternative measures to identify IT skills, such as using the content of each job description, but those alternatives do not qualitatively change the results of our analysis. As with data analytics skills, we construct the level of IT labor for each firm in each year and match it to the firm's performance measures. This measurement of IT labor has been verified to represent the quantity of IT inputs in each firm and can perform better than the traditional capital-based investment measure of IT in the productivity framework (Tambe and Hitt 2013b, Wu et al. 2018). As we do with data analytics, we aggregate employees with IT skills in each firm to measure the overall IT labor after adjusting for the sampling rate. Our primary results are based on classifications where IT and data analytics skills can overlap, but our results are robust to a narrower classification where workers can be counted as having IT skills or data analytics skills but not both.

Production Inputs and Performance

We use the Compustat Industrial Annual files from 1988 to 2013 to measure other production inputs such as physical assets, employees, and production outputs, sales. Materials are calculated as sales minus the sum of operating income and labor expense. If labor expense is not available, we estimate it by multiplying the total number of employees by the industry average wage. These techniques are consistent with approaches used in the R&D and IT-productivity literature. The summary statistics for the financial variables, data analytics, and patent-related variables are shown in Table 2, and the correlation table is shown in Table 3.

Figure 2. (Color online) Average Data Analytics Skills Over Time



Note. The y-axis shows the average percentage of employees with data analytics skills, and the x-axis indicates years.

Empirical Methods

Our primary analysis tests for complementarities between our data analytics measure and various measures of innovation: process improvement and general innovative practices from surveys, and diverse recombination and novelty from patent activity. Two types of statistical tests have been developed to assess the existence of such complementarities (Milgrom and Roberts 1990, Arora and Gambardella 1994, Brynjolfsson and Milgrom 2013): correlations (adoption or demand equations) and performance differences (productivity equations). The correlation test, which can be estimated in the form of conditional correlations or reduced-form demand equations, determines whether a cluster of practices is more likely to be adopted jointly rather than separately. Although the demand equations have the advantage that they are relatively simple and provide the greatest power when firms are optimally matching complementary practices, they have the disadvantage that their simplicity makes them vulnerable to unobserved heterogeneity. In addition, such an analysis will tend to understate the strength of complements if not all firms are endowed with or able to change to the optimal

Table 2. Summary Statistics

Variables	(1) Observations	(2) Mean	(3) Standard deviation	(4) Minimum	(5) Maximum
Summary statistics used to analyze patent data					
ln(*Sales*)	25,635	6.249	2.222	−3.588	13.00
ln(*Materials*)	25,635	5.772	2.143	−2.752	12.74
ln(*Capital*)	25,635	5.540	2.491	−2.162	13.13
ln(*IT labor*)	25,635	4.071	1.919	0	11.60
ln(*Other labor*)	25,635	7.392	2.423	0.00896	14.60
ln(*Data*)	25,635	3.406	3.682	0	11.51
ln(*Diversity*)	25,635	0.256	0.146	0	0.693
ln(*Novelty*)	25,635	0.00923	0.00698	0	0.0938
ln(*Total Neighbor Data*)	20,976	6.915	2.452	0	11.80
ln(*Total Neighbor ERP*)	20,976	1.864	1.429	0	7.192
ln(*Total Neighbor HCM*)	20,976	1.106	1.207	0	5.493
Summary statistics used to analyze survey data					
ln(*Sales*)	2,736	7.003	1.581	0.902	12.218
ln(*Materials*)	2,736	6.510	1.606	−0.317	12.502
ln(*Capital*)	2,736	6.148	1.948	−1.187	12.368
ln(*IT labor*)	2,736	4.653	3.128	0	11.232
ln(*Other labor*)	2,736	8.361	1.680	0	13.123
ln(*Data*)	2,736	4.757	3.937	0	11.14188
ln(*Total Neighbor Data*)	1,966	8.081	1.731	0.693	11.518
ln(*Total Neighbor ERP*)	1,966	1.972	1.637	0	18.563
ln(*Total Neighbor HCM*)	1,966	9.168	16.550	0	167

Notes. Sales, materials, and capital are measured in millions. We add one to actual values of the employee and patent related variables to avoid the possibilities of taking a natural logarithm of 0.

match of the complements. An alternative approach to examining complementarities is to measure whether there are performance differences between individual firms that all adopted complementary practices and firms that adopted only some of the complements. These metrics have the advantage of a direct tie to a relevant firm outcome (performance), and the effects are statistically powerful if not all firms have found the optimal match, which is likely in a situation where firms are implementing new business practices. Over time, as the complementary system diffuses to other firms, the correlation would increase, but the productivity premium would decrease because it no longer differentiates firms when the system becomes widespread. Thus, complementarity results, especially the productivity test, may reasonably capture individual benefits of investments by firms but may understate other benefits to consumers as those benefits are passed on through competition.

We take the view that firm capabilities for pursuing specific innovative practices are quasi-fixed, at least relative to data analytics capabilities, since it is difficult to change them because of organizational inertia, path dependency, culture, norms, and prior strategic choices (Hannan and Freeman 1984, Lam 2005). The rapid diffusion of analytics technology and

Table 3. Correlations

	1	2	3	4	5	6	7	8	9	10	11	12	13
1. ln(*Sales*)	1												
2. ln(*Materials*)	0.970	1											
3. ln(*Capital*)	0.921	0.908	1										
4. ln(*IT labor*)	0.517	0.515	0.496	1									
5. ln(*Other labor*)	0.840	0.803	0.796	0.692	1								
6. ln(*Data*)	0.446	0.440	0.405	0.374	0.479	1							
7. ln(*Diversity*)	−0.0359	−0.0123	−0.0328	−0.0273	−0.0365	0.107	1						
8. ln(*Novelty*)	−0.161	−0.131	−0.140	−0.0388	−0.100	−0.000644	0.538	1					
9. ln(*Total Neighbor Data*)	0.511	0.504	0.440	0.292	0.436	0.423	0.144	−0.0243	1				
10. ln(*Total Neighbor ERP*)	0.183	0.199	0.148	−0.0280	0.0300	0.185	0.193	−0.0590	0.506	1			
11. ln(*Total Neighbor HCM*)	0.476	0.479	0.404	0.305	0.365	0.406	0.193	−0.0218	0.727	0.552	1		
12. *Process*	0.105	0.0992	0.0474	0.0373	0.114	−0.0171	−0.0438	0.00674	0.00577	−0.0566	−0.0207		
13. *Innovation*	0.0646	0.0783	0.0500	−0.0345	0.0955	0.00381	−0.299	−0.190	−0.202	−0.0594	−0.162	0.165	1

an exogenous shift in the amount of data available to firms enable us to observe the effects of data analytics on different firm practices related to innovation, even those that are not optimally configured to benefit from analytics capabilities.

Correlations Between Innovation and Data Analytics

According to our theory, we expect data analytics to be positively correlated with process improvement practices (H1) as well as innovation created by combining diverse technologies (H2) because these practices can be enhanced by greater search capabilities and increased capability for processing information. We expect that data analytics has a weaker correlation with novel technology development than the corresponding correlation with diverse recombination or process improvements (H3), and it is possible that there is no measurable relationship at all.

Here and throughout, we will use italics to designate specific measures. For our analysis of survey data, we match a single cross-section of survey data that provides process and general innovation measures (*Process*, *Innovation*), to a panel of data analytics skills (*Data*), financial (*Sales*), labor supply drivers, time dummies (y_t), and industry controls (*Industry*). Labor supply instruments are described in detail in the identification section.

$$\begin{aligned}\ln(Data)_{it} = \beta_0 &+ \beta_1 Process_i + \beta_2 Innovation_i \\ &+ \beta_3 \ln(Sales)_{it} + y_t + Industry_k \\ &+ \sum_l \gamma_l LaborSupplyInstruments_l + \epsilon_{it} \qquad (5)\end{aligned}$$

We also have a parallel equation for our patent data; the primary difference for empirical analysis between the survey- and patent-based measurement is that we have panel data on two different metrics of innovation (*Diversity*, *Novelty*). These patent-related metrics are time varying, which allow the estimation to include firm fixed effects (γ_i).

$$\begin{aligned}\ln(Data)_{it} = \beta_0 &+ \beta_1 \ln(Diversity)_{it} + \beta_2 \ln(Novelty)_{it} \\ &+ \beta_3 \ln(Sales)_{it} + y_t + \gamma_i \\ &+ \sum_l \gamma_l LaborSupplyInstruments_l + \epsilon_{it} \qquad (6)\end{aligned}$$

These equations can be interpreted as reduced-form demand equations to the extent that the "price" of data analytics skills is varying and the innovative inputs are quasi-fixed relative to data analytics skills.[8] We also introduce instrumental variables that reflect the availability of these skills in the labor market (see the identification section). Thus, Equation (5) and (6) can also be interpreted as the first-stage regression in two-stage least squares (2SLS).

Productivity Benefits of Combining Data Analytics and Innovation

The productivity test examines whether the hypothesized system of complements is more productive if adopted together or separately and is typically implemented by including the direct effects as well as the interaction effects in a productivity or performance model. A complementarities relationship is consistent with a statistically significant interaction term but can also be estimated by sample comparisons that would also imply increasing benefits of one factor in the presence of increasing levels of another complement (see, e.g., Brynjolfsson and Milgrom 2013).

We use the multifactor productivity framework, specifically the Cobb–Douglas production function, to relate firm output such as sales or value-added to various firm inputs such as labor, capital, and materials after controlling for industry (or firms in firm-level fixed effects) and temporal variations. The residual of this equation can be interpreted as the multifactor productivity, and we are interested in exploring whether data analytics and its interaction with different types of innovation are related to this measure of productivity. Specifically, we relate sales to production inputs such as materials expense (*Materials*), physical capital stock (*Capital*), and labor (*IT labor, Data, Other labor*), as well as controls for firm fixed effects (γ_i) and time (y_t). The sales-based specification is more flexible, but results are similar if we adopted a value-added specification (results not shown). The firm fixed effects (or industry controls at the 1.5 digit SIC industry classification) address firm- or industry-specific unobserved heterogeneity, and the year controls address temporal productivity shocks and measurement error in price deflators. Our panel analysis includes the performance and labor metrics on a 26-year panel from 1988 to 2013.

We add to this base specification our measures of the two survey-based firm practices—process improvements (*Process*) and general innovation practices (*Innovation*)—and their interactions with data analytics to evaluate complementarities.

$$\begin{aligned}\ln(Sales)_{it} = \beta_0 &+ \beta_1 \ln(Materials)_{it} + \beta_2 \ln(Capital)_{it} \\ &+ \beta_3 \ln(Other\ labor)_{it} + \beta_4 \ln(IT\ labor)_{it} \\ &+ \beta_5 \ln(Data)_{it} + \beta_6 Process_i + \beta_7 Innovation_i \\ &+ \beta_8 Data_{it} \times Process_i \\ &+ \beta_9 Data_{it} \times Innovation_i + y_t + \gamma_i \\ &+ Controls + \epsilon_{it} \qquad (7)\end{aligned}$$

We also estimate a similar model using our patent-derived measures of diverse recombination (*Diversity*) and novel innovation (*Novelty*) to examine their complementary relationships with data analytics after controlling for patent stock.[9]

$$\begin{aligned}\ln(Sales)_{it} = {} & \beta_0 + \beta_1 \ln(Materials)_{it} + \beta_2 \ln(Capital)_{it} \\ & + \beta_3 \ln(Other\,labor)_{it} + \beta_4 \ln(IT\,labor)_{it} \\ & + \beta_5 \ln(Data)_{it} + \beta_6 \ln(Diversity)_{it} \\ & + \beta_7 \ln(Novelty)_{it} + \beta_8 Data_{it} \times Diversity_{it} \\ & + \beta_9 Data_{it} \times Novelty_{it} + y_t + \gamma_i \\ & + Controls + \epsilon_{it}\end{aligned} \tag{8}$$

Identification

Complementarities approaches are naturally robust to some types of endogeneity and reverse causality problems because they are ultimately about matching two or more investment decisions rather than the effectiveness of the decisions themselves. While there are many explanations for why an investment might be correlated with unobserved shocks, it is hard to explain why the presence or absence of concurrent investment outperforms investment in one factor but not another. In addition, the correlation test and the productivity test for complementarities are subject to different sources of bias, so consistent results provide greater confidence that the results are not primarily due to econometric issues. Lastly, we explore multiple complementarities; the presence of one without the other can also help rule out simple cases of selection biases.

Nonetheless there are at least two endogenous factors that could lead to a positive bias in our estimate. First, high-performing firms can have slack resources that can be deployed to invest in data analytics acquisition, which could lead to a reverse causality between performance and data analytics. By itself this is not a problem in interpreting our complementarities result but could introduce bias if this tendency was higher in firms that engaged in certain types of innovation—for instance, firms that have created novel innovations. Second, there could be omitted variables, such as higher management quality also attracting higher-quality labor, that lead to an upward bias. We address these problems in several ways. First, our results using patent data are based on fixed effects models where we control for time-invariant firm characteristics. Second, we consider multiple forms of innovation and find that there are complementarities with some and not others, ruling out simple stories related to omitted variables or selection biases on the general ability of firms to innovate. Finally, we treat data analytics skills as endogenous and use three sets of instruments to correct for potential biases: (1) data analytics skills in a firm's existing hiring network; (2) a neighbor's adoption of an enterprise resource planning (ERP)[10] system and the total number of neighbors that started to use ERP systems; and (3) a neighbor's adoption of a human capital management (HCM) system and the total number of neighbors that started to use HCM. All of these instruments are derived from a firm's immediate hiring network, which can be used as a proxy to measure the ease of access to existing data analytics talent and therefore the "cost" of acquiring analytics skills. This hiring network is constructed by examining employee flows across firms using techniques similar to those in Wu et al. (2018). This approach has the advantage of meeting the first-stage correlation requirements but is inherently more immune to many common shocks because the firms matched by labor flows are diverse both in industry and geography that are subject to different shocks.

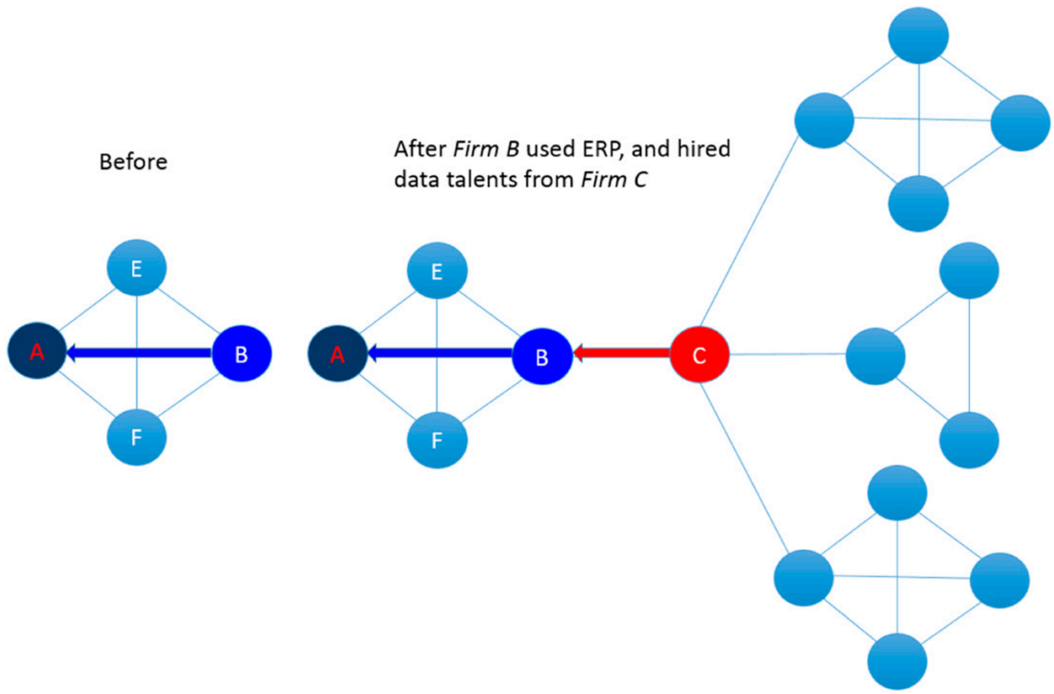
In the network, firms are represented by nodes, and firm-to-firm employee movements are represented by edges. We identify a directed edge between Firm A and Firm B if A hired one or more employees from B in the previous five years.[11] The weight of that directed edge is the number of employees that A has hired from B. The first instrumental variable is a Hausman-type instrumental variable, based on the neighboring firm's existing data talent. This variable measures the potential labor pool accessible to the firm through employee mobility, which reflects the cost of acquiring additional analytics talent. At the same time, the pool of data analytics skills from the neighboring firms is only the potential access a firm could have, and thus it should not affect the firm's own performance. The only time that the talent pool could affect performance of the focal firm is when the firm hires from a neighboring firm, which we can capture through the firm's own data analytics skills. The firm fixed-effect model also captures the effect from the change in data analytics skills between years and can thus account for the talent flow into the firm. Accordingly, the potential for productivity spillovers through labor movement would not invalidate our instruments.

The second and third instrumental variables are derived from measures of the adoption of ERP systems broadly and the HCM module by firms in the same labor pool. These types of systems are likely to be complementary to the use of data analytics skills, and thus can change the overall access to data analytics skills; the access is expanded if new employees become available in the talent pool, but it can also be reduced if neighboring firms compete for the same skills. In addition, the ERP adoption event may have an especially large effect on the availability of employees because of a transition from the implementation phase to the system being fully operational, which may free up the labor involved in the setup. We focus specifically on the point in time a firm implements a system (rather than simply licenses one, which could be several years earlier) to prevent common industry-level shocks from potentially generating an increased demand for these systems in nearby firms as well as a direct shock to productivity in the focal firm (Hitt and Brynjolfsson 1996, Hitt et al. 2002). Thus, while a neighboring firm's adoption of ERP should not directly affect the focal firm's productivity, it should have an impact on the focal firm's access to data analytics–related talent. To illustrate this, we show in Figure 3 the impact on the access to data talent of the focal firm (Firm A) when its neighbor (Firm B), from which Firm A hires people, adopted an ERP system. Firm B now has a need to hire employees with ERP experience as well as experience working with the process changes that often accompany ERP implementations, and starts to hire from new firms, such as Firm C. Not only has Firm B's hiring pool expanded, at the same time, this network change also propagates to Firm A's access to data analytics–related talent. As shown in Figure 3, Firm A still hires from the same three firms, but its access to data analytics has expanded because of Firm B's new hires that it acquired from Firm C. On the other hand, if Firm B hired from its neighboring firms (Firms E and F) directly, it would reduce Firm A's potential access to data analytics skills. Thus, Firm B's ERP adoption induces a change in Firm A's access to talent even though Firm A has not changed its hiring practices. We can then use Firm B's ERP adoption event as an instrument for Firm A's access to data analytics skills.

We have two available measures of ERP and HCM adoption. First, we can infer ERP adoption by examining whether employees in a firm have job titles or responsibility descriptions in their résumés that suggest the use of ERP (e.g., employees stating in their past job description that they "used an ERP system to conduct inventory control"). We also have access to when firms purchased and adopted an HCM module from a major ERP vendor. This is used to construct a binary variable to represent the firm's HCM implementation status, which is 0 in the years before the firm started using the system and 1 afterward.

Another potential source of endogeneity is that innovation practices may also be endogenous. Here, we rely on the assumption that organizational practices are often quasi-fixed (Milgrom and Roberts 1990,

Figure 3. (Color online) Illustration of Instrumental Variables



Notes. Firms are represented by circular nodes, and firm-to-firm employee movements are represented by edges. The graph on the left displays the hiring network of Firm A. We identify a directed edge between Firm A and Firm B if Firm A hired one or more employees from Firm B in the previous 5 years. Assume that each firm on the left graph has 10 employees with data analytics skills, and accordingly Firm A has access to 30 employees with data talent. The graph on the right shows what happens to Firm A's access to data talents when Firm B, its network neighbor, uses an ERP system. When Firm B hires 5 more data analytics employees from Firm C after ERP adoption; Firm A still hires from the same 3 firms, but its access to data analytics skills has expanded because of Firm B's new hires that it acquired from Firm C. Now Firm A can access 35 people with data talent. On the other hand, if Firm B hired from its neighboring firms (Firm E and Firm F) directly after ERP adoption, it would reduce Firm A's potential access to data analytics skills. Thus, Firm B's ERP adoption induces a change in Firm A's access to talent even though Firm A has not changed its own hiring practices. We can then use Firm B's ERP adoption event as an instrument for Firm A's access to data analytics skills.

Brynjolfsson and Hitt 1996, Levy and Murnane 1996), at least relative to data analytics investments. The quasi-fixity of our innovation practices measures may be especially reasonable given the large corpus of literature that has documented the difficulty firms face in changing their innovation practices and the nature of their innovation portfolios, because it requires firms to substantially alter practices in many areas including finance, talent acquisition, monitoring policy, incentive pay, and reporting relationships (Nagji and Tuff 2012, Wessel and Christensen 2012). We empirically verify the extent to which the innovation characteristics are quasi-fixed. Overall, the standard deviations of the innovation characteristics (diversity and novelty) within firms are only a quarter of their respective means,[12] suggesting that it is difficult for firms to change their innovation practices in the short term. Thus, our regression results can be interpreted as assessing whether preexisting firm differences in innovation practices influence the benefits of acquiring data analytics capabilities. However, since we do have time-series measures for the patent-related constructs, we include them as control variables to remove any direct effect of these variables to allow for a more straightforward interpretation of our interaction term estimates.

Finally, we address general endogeneity of production inputs with respect to output using panel-based instrumental variables methods (the SYS-GMM estimator of Blundell and Bond 1998) that were developed to specifically address problems of endogeneity in microlevel productivity analysis. The combination of instrumental variables and SYS-GMM in a complementarities framework should substantially reduce the potential biases in our results by causality problems. However, we acknowledge that our archival data may not be able to definitively address causality, which is extremely difficult when examining the economic impact in a broad sector of economy as is the case in our study.

Results

Descriptive Statistics

The survey questions used to measure organizational practices are shown in Table 1. Table 2 shows the summary statistics for the survey-based analysis and the patent-related analysis, and Table 3 shows the pairwise correlation table among the variables of interest. Although innovation and process improvement practices are correlated, the correlation coefficient is only 0.165, suggesting that we should observe a variety of different combinations of innovative practices in our data. Figure 1 shows the distribution of firms along the process and innovation axes. Firms are evenly distributed in the four quadrants. We also find that the correlation between recombination diversity and novelty as measured in the patent data are 0.538, suggesting a variety of combinations are used in practice and that the measures are distinct, even though they are derived from the same underlying of data.

Evidence on Data Analytics and Survey-based Firm Practices

First, we explore whether process improvement practices or general innovation practices lead to a higher demand for data analytics capabilities in Table 4. We first estimate the model using ordinary least squares (OLS) with standard errors clustered at the firm level. We find that a one-standard-deviation increase in process improvement practices is associated with a 2.22% increase in hiring employees with data analytics capabilities (column (1)), suggesting that the two are complements as observed in our data. However, the same increase in innovation practices does not have a statistically significant effect on the demand for data analytics (column (2)). Column (3) uses both process and innovation variables in the same model and shows similar results as earlier columns. The three models include instrumental variables (data capabilities, ERP and HCM adoptions in neighboring firms) as controls and thus can be interpreted as a first-stage analysis in 2SLS. The R^2 across all models is above 0.60, and the associated concentration parameter is well above the criteria for the weak instrument test. Detailed first-stage statistics are in the note section of Table 5. The point estimates in the first stage regression for the instruments appear to vary substantially, likely because of correlations among the instruments—this is not an issue for their use as instrumental variables, but we are reluctant to interpret their magnitude or directional effect individually.

Next, we examine the presence of complementarities using the production function framework and all models use clustered standard errors at the firm level in Table 5.[13] If data analytics and process improvement practices are complementary, firms that possess the complementary system should experience a greater return than firms that have only one of the two conditions. We first examine the marginal effect of data analytics in the production function framework (column (1)). We observe that a 1% increase in data analytics skills is associated with a 0.0273% increase in sales. We also include the process and innovation practice variables on their own, but their effects are statistically insignificant (column (2)). It is possible that we lack the statistical power to provide precise estimates, but it is also possible that certain business practices require the presence of organizational complements to be valuable. To examine the presence of complementarities, we show the pooled OLS result in column (3); the interaction between data analytics and process improvement practices is positive after controlling for industry and year dummies.

Table 4. Relationship Between Data Analytics Skills and Organizational Characteristics

	OLS		
Model	(1)	(2)	(3)
Process	0.0222***		0.0213**
	(0.00846)		(0.00831)
Innovation		0.0102	0.00549
		(0.00787)	(0.00733)
ln(*Sales*)	0.00130	0.00113	0.000536
	(0.00419)	(0.00460)	(0.00454)
ln(*Total Neighbor Data*)	−0.0385	−0.0505	−0.0407
	(0.0412)	(0.0402)	(0.0413)
ln(*Total Neighbor ERP*)	−0.00149	−0.00162	−0.00154
	(0.00273)	(0.00269)	(0.00272)
ln(*Total Neighbor HCM*)	0.0332***	0.0338***	0.0332***
	(0.00794)	(0.00831)	(0.00793)
Lagged ln(*Total Neighbor Data*)	0.0222	0.0344	0.0250
	(0.0388)	(0.0372)	(0.0388)
Lagged ln(*Total Neighbor ERP*)	−0.00175	−0.00268	−0.00171
	(0.00230)	(0.00233)	(0.00233)
Lagged ln(*Total Neighbor HCM*)	−0.0227***	−0.0231***	−0.0228***
	(0.00786)	(0.00814)	(0.00786)
Observations	1,966	1,966	1,966
R^2	0.621	0.617	0.621
Number of firms	145	145	145
Industry	Yes	Yes	Yes
Year	Yes	Yes	Yes

Notes. The dependent variable is ln(*Data*). The survey-based regressions (columns (1)–(3)) use OLS due to the cross-sectional nature of the surveys. Sales is measured in millions. Robust (clustered by firm) standard errors are reported in parentheses.

$^{**}p < 0.05$; $^{***}p < 0.01$.

On average, the interaction term implies that a one-standard-deviation increase in the use of data analytics is associated with a 5.05% ($p < 0.05$) increase in productivity in firms that are one standard deviation above the mean in the use of process improvement practices (that is, they are complementary). We also observe a similar effect in the firm-fixed-effect model in column (4). The fixed-effect model drops the process improvement and innovation practices variables because they are cross-sectional. Our models include IT labor, suggesting that the variation in productivity from the data-process complementarity is not due to ordinary returns to IT investments.[14] On the other hand, the interaction between data analytics and general innovation practices is mixed. The interaction in pooled OLS is negative but only statistically significant at the 10% level; it becomes statistically insignificant in the firm-fixed-effect model in column (4). In column (5), we use instrumental variables to examine complementarities in a production function. The first-stage analysis using these instrumental variables is shown in Table 4. The first-stage F-statistics across all models are comfortably greater than 40, and the Sargan/Hansen overidentification criteria are also satisfied (see note section of Table 5). The 2SLS results are consistent with the OLS and fixed effects; the interaction between data analytics and innovation practices is statistically insignificant, whereas the interaction between data analytics and process improvement practices is positive ($\beta = 0.151$, $p < 0.01$). We then use the SYS-GMM approach with the same set of external instrumental variables to estimate a two-step GMM model with robust standard errors. Using three-period lags of the production input variables to control for the potential endogenous factors that arise from production inputs such as materials, capital, and labor, we find that data analytics positively interacts with process improvement practices ($\beta = 0.073$, $p < 0.05$) but not with innovation practices (column (6)).

Overall, we find support for the hypothesis that data analytics supports process innovation (H1). These results are consistent for both the productivity and demand models and across various methods (OLS, fixed effects [FE], 2SLS, GMM) for productivity modeling. Our results are inconclusive about the relationship between data analytics and general innovation practices as measured in the survey. This could be that the general innovation is a mix of different types of innovation and only some of them are complementary to analytics. We thus explore these relationships in more detail in the patent data in the following section.

Patent Data Analysis: Diversity and Novelty

Table 6 shows the correlations between data analytics and patents that indicate innovation through diverse recombination (*Diversity*), and between data analytics and patents that indicate innovation through novel innovation (*Novelty*). Because we have time series data on these constructs, firm-level fixed effects are applied across the three models in Table 6, and standard errors are clustered at the firm level. We show a positive correlation between data analytics and diversity ($\beta = 0.700$, $p < 0.01$ in column (1)). On the other hand, novelty is not statistically significantly correlated with data analytics. As with the prior analysis using the survey data, these models include our instruments and thus can be considered as first-stage models for 2SLS using the productivity framework. Overall, the values for R^2 are about 0.70 and the concentration parameter is well above the threshold for passing the weak instrument test. Detailed first-stage statistics are shown in the note section of Table 7. Using an alternative novelty metric based on patent class shows consistent results (Online Appendix C). The results are also statistically insignificant if we attempt to correct for measurement error using alternative measurements of novelty to

Table 5. Data Analytics Skills, Firm Practices, and Productivity

Model	(1) FE	(2) OLS	(3) OLS	(4) FE	(5) 2SLS	(6) GMM/IV
ln(*Materials*)	0.672***	0.674***	0.654***	0.637***	0.659***	0.605***
	(0.0402)	(0.0396)	(0.00954)	(0.0801)	(0.0149)	(0.0525)
ln(*Capital*)	0.188***	0.187***	0.139***	0.253***	0.0975***	0.168***
	(0.0284)	(0.0279)	(0.00921)	(0.0553)	(0.0191)	(0.0356)
ln(*Other labor*)	0.107***	0.104***	0.204***	0.0871**	0.284***	0.205***
	(0.0301)	(0.0289)	(0.0111)	(0.0379)	(0.0202)	(0.0404)
ln(*IT labor*)	0.000830	0.00242	–0.00448	0.0118	0.0175***	–0.00703
	(0.00694)	(0.00680)	(0.00297)	(0.00740)	(0.00292)	(0.00892)
ln(*Data*)	0.0273*	0.0274*	–0.00471	0.00571	–0.0834***	0.00206
	(0.0141)	(0.0144)	(0.00849)	(0.0220)	(0.0275)	(0.0253)
Process		0.0111	–0.00388			–0.0159
		(0.0218)	(0.00806)			(0.0176)
Innovation		0.0197	0.00430			–0.0212
		(0.0237)	(0.00853)			(0.0206)
Data × *Process*			0.0505**	0.118***	0.151***	0.0731**
			(0.0217)	(0.0398)	(0.0281)	(0.0310)
Data × *Innovation*			–0.0743*	–0.0275	–0.0451	–0.0156
			(0.0389)	(0.0746)	(0.0500)	(0.0670)
Observations	2,736	2,736	2,736	2,736	1,966	1,966
R^2	0.948	0.948	0.954	0.872	0.887	0.887
Number of firms	163	163	163	163	145	145
Industry	—	Yes	Yes	—	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable is ln(*Sales*). Column (1) presents the baseline estimates with firm fixed effects. Columns (2) and (3) show the pooled OLS estimates. Column (4) shows the firm-fixed effect version of the estimate in column (3). Columns (5) and (6) show the 2SLS and GMM estimates using 6 external instruments. Six instrumental variables are used: (1) total number of employees with data analytics skills in neighboring firms, (2) the associated one-period lagged, (3) total number of neighbors that adopted an enterprise resource planning (ERP), (4) the associated one period lag, (5) total neighbors that adopted Human Capital Management (HCM) systems, and (6) and the associated one-period lag. The first-stage diagnostics are satisfied. The Sargan-Hansen overidentification test cannot reject the null that the overidentification is valid ($\chi^2 = 2.645$, $p < 0.267$). The Cragg–Donald Wald F-statistics for first-stage regression is 42.41, above the recommended value of 10. We use the Blundell and Bond (1998) SYS-GMM estimator that derives additional instruments using the lagged level and differences from production inputs. We use three-period lags of the endogenous variables as instruments in addition to the external instruments used in 2SLS. We use the two-step estimation procedure. The Arellano–Bond test for AR(2) in first differences cannot reject the null that there are serial correlations ($p < 0.118$). Neither the Hansen J statistic of overidentification ($p = 0.134$) nor the difference-in-Hansen test of the system GMM instruments ($p = 0.239$) rejects the null that the instruments are uncorrelated with the error term. We also test other lags and find they do not qualitatively change our results. All interactions are formed from centered variables. Sales, materials and capital are measured in millions. Robust (clustered by firm) standard errors are reported in parentheses.

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

instrument for the word-age–based measurement (results not shown).

We show the productivity analysis in Table 7.[15] All models include firm fixed effects and controls for patent stock. First, we present our regression results without the interaction and find that data analytics has a positive effect in the production function framework. The direct effects of diversity and novelty are not statistically significant, which is consistent with earlier innovation literature that uses similar measures, and suggests that certain technology practices or other business practices require the presence of necessary complements to be valuable (Huselid 1995, Cappelli and Neumark 2001, Bresnahan et al. 2002, Aral et al. 2012). We then show that data analytics and diverse recombination are complements in the production function: the interaction between data analytics and diversity is positive ($\beta = 0.010$, $p < 0.01$). However, the interaction with novelty is statistically insignificant (column (3)).

Next, we use instrumental variables in 2SLS and GMM to ensure that our results are robust for certain types of reverse causality and selection biases. We treat data analytics as endogenous and use the same set of instrumental variables as the first-stage analysis in Table 6. The sample size for columns (4) and (5) is smaller because of incomplete data on instruments. The first-stage F-statistics are comfortably greater than

Table 6. Relationship Between Data Analytics Skills and Innovation

	FE		
Model	(1)	(2)	(3)
ln(*Diversity*)	0.700***		0.815**
	(0.268)		(0.331)
ln(*Novelty*)		6.671	–3.945
		(5.702)	(6.973)
ln(*Sales*)	0.839***	0.852***	0.838***
	(0.0690)	(0.0690)	(0.0691)
ln(*Total Neighbor Data*)	0.348**	0.346**	0.346**
	(0.154)	(0.154)	(0.154)
ln(*Total Neighbor ERP*)	0.273	0.269	0.272
	(0.236)	(0.237)	(0.236)
ln(*Total Neighbor HCM*)	–0.0445	–0.0443	–0.0433
	(0.106)	(0.106)	(0.106)
Lagged ln(*Total Neighbor Data*)	–0.0937	–0.0907	–0.0917
	(0.136)	(0.136)	(0.136)
Lagged ln(*Total Neighbor ERP*)	–0.216	–0.213	–0.219
	(0.231)	(0.231)	(0.231)
Lagged ln(*Total Neighbor HCM*)	0.0324	0.0306	0.0323
	(0.100)	(0.100)	(0.100)
Observations	18,716	18,716	18,716
R^2	0.700	0.700	0.700
Number of firms	1,934	1,934	1,934
Year	Yes	Yes	Yes

Notes. The dependent variable is ln(*Data*). The patent-based regressions (columns (1)–(3)) use fixed effect models as diversity and novelty are both time varying. All models also include firm's patent stock as a control. We also test the yearly number of patents filed by the firm, which yields similar effect. Sales is measured in millions. Robust (clustered by firm) standard errors are reported in parentheses.

$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

20, and the Sargan/Hansen overidentification criteria are also satisfied. Consistent with our earlier analysis, 2SLS shows that data analytics and diverse recombination are complementary—that is, the marginal return on investment in analytics skills is greater in firms engaged in innovation by combining diverse sets of knowledge. However, the interactions between data analytics and novel innovation are statistically insignificant. Next, we use dynamic SYS-GMM to address other forms of reverse causality between production inputs and outputs that have been argued to affect production functions (e.g., sales shocks leading to greater investment), which could create biases in all coefficients. Using both internal lags and external instrumental variables used in 2SLS, we find consistent evidence that data analytics complements diverse recombination, but the interaction with novelty is not statistically significant (column (5)). We then examine whether IT labor has a similar interaction effect to data analytics skills. After including the interactions with IT labor, we continue to find that the interaction between data analytics and diverse recombination is positive ($\beta = 0.0115$, $p < 0.01$), while the interaction between data analytics and novelty is not (column (6)). This shows that effects we observe are unique to the impact of data analytics and are not simply due to the impact of general IT capabilities. We also include the interactions with process and innovation variables from the survey in column (7). While the sample is substantially reduced after merging the survey with the existing sample, all of our core results remain the same. Lastly, we create disjointed labor inputs of IT and data labor variables such that an employee can have either data analytics skills or IT skills, but not both. Using nonoverlapping labor inputs, we continue to observe that data analytics complement diverse recombination but not novelty in column (8) (also see Online Appendix C), suggesting that the potential double counting of labor inputs has minimal effect on our estimates.

Finally, although we include year dummies to address issues related to combining two data sources to create a more comprehensive measure of data analytics skills, we estimate our models separately for each data source to ensure that our results are not driven by the data merge. Table 8 shows that splitting the samples does not change our core results.

As a final robustness check, we address whether that the lack of a relationship with novelty is due to weak measurement by instrumenting the word-age measure with alternative measurements of novelty—this should reduce bias from measurement errors if the errors in different measurements are uncorrelated. These estimates show narrower confidence intervals, but they are still not statistically significant (results not shown). The use of several distinct novelty measures (word-age, new patent classes, and an LDA-based measure) along with the 2SLS estimates provides some confidence that our lack of findings related to novelty is not primarily driven by measurement problems or lack of statistical power.

Overall, these results suggest that data analytics is complementary to innovation related to process improvement but not general innovation practices (as predicted by H1). Data analytics is also shown to be complementary to diverse recombination (as H2 predicted). However, we find complementarities with novel innovation are much weaker than the complementarities with diverse recombination (in line with H3). Our predictions are supported for both demand and productivity models, and are robust to various instrumental variables methods applied to the productivity equation. Viewing these results collectively also helps alleviate certain simple scenarios of selection and omitted variable biases. It is difficult to envision a selection process that would have data analytics selectively complement process improvement but not general innovation practices, and have data analytics selectively complement diverse

Table 7. Data Analytics Skills, Innovation, and Productivity

Model	(1) FE	(2) FE	(3) FE	(4) 2SLS	(5) GMM/IV	(6) FE	(7) FE	(8) FE
ln(*Materials*)	0.610***	0.610***	0.610***	0.593***	0.697***	0.610***	0.773***	0.610***
	(0.0145)	(0.0145)	(0.0145)	(0.0149)	(0.0317)	(0.0145)	(0.0759)	(0.0145)
ln(*Capital*)	0.0941***	0.0940***	0.0929***	0.0875***	0.104***	0.0931***	0.132**	0.0924***
	(0.0110)	(0.0111)	(0.0110)	(0.0112)	(0.0291)	(0.0111)	(0.0617)	(0.0111)
ln(*Other labor*)	0.267***	0.267***	0.268***	0.238***	0.219***	0.267***	0.171***	0.268***
	(0.0119)	(0.0118)	(0.0118)	(0.0236)	(0.0303)	(0.0118)	(0.0514)	(0.0118)
ln(*IT labor*)	0.00949***	0.00952***	0.00944***	0.0462**	0.000737	0.00928***	0.00912	0.0112***
	(0.00239)	(0.00238)	(0.00238)	(0.0201)	(0.00974)	(0.00258)	(0.00930)	(0.00253)
ln(*Data*)	0.00161**	0.00163**	0.00149*	0.0269	0.00274	0.00154*	0.0237	0.00800***
	(0.000787)	(0.000786)	(0.000803)	(0.0275)	(0.00224)	(0.000806)	(0.0440)	(0.00163)
ln(*Diversity*)		–0.00134	–0.00852	–0.0726	–0.0655	–0.00793	0.266	–0.00823
		(0.0250)	(0.0253)	(0.0516)	(0.0444)	(0.0253)	(0.228)	(0.0253)
ln(*Novelty*)		–0.808	–0.786	0.355	–1.684	–0.785	0.920	–0.791
		(0.556)	(0.558)	(1.026)	(1.052)	(0.558)	(4.343)	(0.559)
Data × *Diversity*			0.0103***	0.0247***	0.0178***	0.0115***	0.0759**	0.00911**
			(0.00234)	(0.00844)	(0.00667)	(0.00238)	(0.0432)	(0.00415)
Data × *Novelty*			–0.00270	–0.0159	0.00305	–0.00281	–0.0312	0.000314
			(0.00276)	(0.0110)	(0.00898)	(0.00284)	(0.0348)	(0.00536)
IT × *Diversity*						–0.00513		–0.00168
						(0.00321)		(0.00428)
IT × *Novelty*						8.75×10^{-5}		0.00339
						(0.00349)		(0.00607)
Data × *Process*							0.0733***	
							(0.0193)	
Data × *Innovation*							0.0818	
							(0.0644)	
Observations	25,635	25,635	25,635	18,716	18,716	25,635	1,099	25,635
R^2	0.927	0.927	0.927	0.895		0.927	0.926	0.927
Number of firms	2,337	2,337	2,337	1,934	1,934	2,337	87	2,337
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable is ln(*Sales*). Columns (1)–(3) and (6)–(8) show the estimates with firm fixed effects. Columns (4) and (5) show the 2SLS and GMM estimates using 6 external instruments. Column (8) uses IT-only, data-only, and other labor such that there is no overlap in the labor inputs. Six instrumental variables are used for data analytics skills: (1) total number of employees with data analytics skills in neighboring firms; (2) one-period lagged values of (1); (3) and (4) total number of neighbors that adopt an enterprise resource planning (ERP) or human capital management (HCM) systems; and (5) and (6) one-period lagged values for ERP or HCM variables in what are calculated for (3) and (4). The first-stage diagnostics are satisfied. The Sargan–Hansen overidentification test cannot reject the null that the overidentification is valid ($\chi^2 = 2.55$, $p < 0.636$). The Cragg–Donald Wald *F*-statistics for first-stage regression is 22.23, above the recommended value of 10. We use the Blundell and Bond (1998) SYS-GMM estimator, which derives additional instruments using the lagged level and differences from production inputs. We use three-period lags of the endogenous variables as instruments in addition to the external instruments used in 2SLS. We use the two-step estimation procedure. The Arellano–Bond test for AR(2) in first differences cannot reject the null that there are serial correlations ($p = 0.207$). Neither the Hansen test of overidentification ($p = 0.271$) nor the difference-in-Hansen test of the system GMM instruments ($p = 0.398$) rejects the null that the instruments are uncorrelated with the error term, ensuring the validity of the instruments used in the GMM estimation. We also test other lags and find they do not qualitatively change our results. Firm's patent stock is also controlled. All interactions are formed from centered variables. Sales, materials, and capital are measured in millions. Robust (clustered by firm) standard errors are reported in parentheses.

$^*p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

recombination but not novel innovation. Our results do not appear to be affected by different treatments of IT labor, adjustments for labor quality, and the use of different measures for innovation.

Conclusion

By using existing data analytics skills as a proxy for data analytics capabilities, we find a significant complementarity between process improvement practices and data analytics capabilities, suggesting that newly available data support decision making on process improvement. We also find that data analytics skills are consistently complementary with innovation that combines a diverse set of existing ideas but do not appear to be complementary with the development of entirely new innovations. Furthermore, our results are related specifically to data analytics but not general IT capabilities.

Table 8. Data Analytics Skills, Innovation, and Productivity: Split Samples

	FE					
	Résumés: 1988–2007			Job reviews: 2008–2013		
Model	(1)	(2)	(3)	(4)	(5)	(6)
ln(*Materials*)	0.591***	0.591***	0.591***	0.712***	0.712***	0.713***
	(0.0148)	(0.0148)	(0.0148)	(0.0209)	(0.0209)	(0.0209)
ln(*Capital*)	0.101***	0.101***	0.0998***	0.108***	0.109***	0.108***
	(0.0109)	(0.0109)	(0.0109)	(0.0158)	(0.0158)	(0.0157)
ln(*Other labor*)	0.264***	0.265***	0.265***	0.115***	0.114***	0.115***
	(0.0122)	(0.0122)	(0.0121)	(0.0141)	(0.0141)	(0.0141)
ln(*IT labor*)	0.00661**	0.00661**	0.00595*	0.0176***	0.0176***	0.0173***
	(0.00324)	(0.00324)	(0.00321)	(0.00251)	(0.00252)	(0.00250)
ln(*Data*)	0.00153**	0.00155**	0.00141*	–0.00169	–0.00163	–0.00237
	(0.000771)	(0.000770)	(0.000772)	(0.00106)	(0.00107)	(0.00172)
ln(*Diversity*)		–0.0214	–0.0337		–0.0449	0.00948
		(0.0229)	(0.0230)		(0.0591)	(0.0572)
ln(*Novelty*)		0.183	0.236		1.595	0.816
		(0.528)	(0.530)		(2.348)	(2.353)
Data × *Diversity*			0.0135***			0.0184***
			(0.00229)			(0.00409)
Data × *Novelty*			–0.00389			–0.0101
			(0.00267)			(0.00868)
Observations	20,976	20,976	20,976	4,659	4,659	4,659
R^2	0.927	0.927	0.928	0.937	0.937	0.937
Number of firms	2,140	2,140	2,140	1,170	1,170	1,170
Year	Yes	Yes	Yes	Yes	Yes	Yes

Notes. The dependent variable is ln(*Sales*). Columns (1)–(6) show the estimates with firm fixed effects. Firm's patent stock is also controlled. All interactions are appropriately centered. Sales, materials, and capital are measured in millions. Robust (clustered by firm) standard errors are reported in parentheses.
$^{*}p < 0.1$; $^{**}p < 0.05$; $^{***}p < 0.01$.

Overall, these results suggest that data analytics capabilities are most strongly associated with innovation that requires either searches across broad knowledge domains or extensive information processing. Innovation where there is either no data (e.g., novel innovation) or where search has limited value (narrow recombination) do not appear to be complementary to data analytics. However, it should be noted that there are no strong theoretical predictions about the relationship between data analytics and novel innovation, and novelty is particularly difficult to measure making it possible that the absence of results in our study is due to a lack of statistical power. However, it is less likely given we find a similar lack of results across multiple distinct measures and methods. It is also possible that it takes a longer time for data analytics to facilitate novel innovation than it does for diverse recombination; thus, detecting any effects on novelty would require a longer time horizon, especially for the later years, when the growth in analytics skills is faster.

These results are consistent with our arguments that there are clear advantages to using existing data to improve existing products and processes. It is similarly consistent with observations that there are challenges in using existing data to generate completely new ideas. These relationships need not be static, however. Technology has been increasing the types of decisions amenable to automation and more types of data are continually being collected or becoming accessible (e.g., speech, image, and video processing) that will cause analytics to expand into new domains. Analytics can still improve the overall innovative processes within firms that could expand innovative output more generally. Improvements in technological diversity in recombination may also raise the value of novel innovation since each new "building block" expands the potential space of recombination in a combinatorial manner. Our results also help to explain why some have claimed a slowdown in the rate of innovation (Bloom et al. 2017), despite many technology advances. Our analysis on data analytics and innovation helps to clarify the processes by which this could occur—that is, some types of innovation (process improvement and diverse recombination) are better supported than others (novel innovation) so that gains in some areas are offset (if only by averaging) with limited benefits in others.

Data analytics and practices in process improvement or diverse recombination act as complements to deliver greater benefits to firms that have these practices in place. This suggests that the scarce resources focused on big data analytics are best directed at innovation involving operational improvements and innovation involving combining existing technologies especially from diverse sources rather than creating truly novel technology. Moreover, investments in the technology and human resources to apply analytics to these types of applications are likely to generate visible payoffs. However, the payoff from using analytics tools to support novel innovation is uncertain because pursuing this approach is more speculative. It is likely that these relationships will continue to evolve as analytics capabilities become more widespread and new capabilities are developed to analyze and act on an ever-expanding array of available information.

Endnotes

[1] The n technical elements can generate potential innovation in 2^n different ways. The $n + 1$th technical element can be combined with all existing n elements in 2^n ways, generating an innovation space of the size 2^{n+1}.

[2] Development of business processes is officially recognized as a protectable form of innovation (see, e.g., https://www.uspto.gov/patents-getting-started/patent-basics/types-patent-applications/utility-patent/patent-business).

[3] The 1976–2006 patent data come from https://sites.google.com/site/patentdataproject/Home. We also refer to multiple data sources, including the earlier version of the data that can be found at http://www.nber.org/patents, USPTO, and patents stored on Google, for the patents before 1976.

[4] A similar aggregation technique can be found in Saunders (2011).

[5] To the extent that such matchings vary systematically across firms, we can address the problem by using fixed-effect models to alleviate the bias from time-invariant firm characteristics.

[6] We search the word "data" in the Occupation Search box provided by https://www.onetonline.org, and obtain a list of job titles ranked according to how well these job titles matched the keyword "data," according to the matching process in https://www.onetonline.org/help/online/search#keyword. A total of 576 primary data–related job titles are generated, with detailed reports on the requisite technology skills and general skills. An additional 212 alternate occupation titles are found from the primary job titles. For example, the occupation "Database Administrators" has several alternate job titles, such as data engineer, data miner, and data center manager. We count a person as having data analytics skills if any of the skills in that job title are related to data.

[7] The F-statistics associated with Chow test $F(3,2336) = 0.62$ ($p < 0.6$). The variables used for merging are *Data*, *Data* × *Diversity*, and *Data* × *Novelty*, as shown in column (3) of Table 7.

[8] A typical demand equation relates the quantity to the prices of a factor and the levels of quasi-fixed factors. Sales can also appear in this equation if the demand system is assumed to arise from cost minimization (since a cost minimization model treats sales as exogenous). In our specification, sales also controls for firm size.

[9] We control for labor quality (education) as a robustness check (Online Appendix C) but omitted it from our main analyses as it does not affect our primary coefficients of interest and is only available for a fraction of the sample.

[10] ERP systems are off-the-shelf software packages that offer a variety of functions including finance, human resources, supply chain management, consumer relationship management, and business intelligence (Hitt and Brynjolfsson 1996). These types of enterprise information systems often produce a large amount of data about firms' business processes, which can affect the demand for data analytics skills that can analyze and make use of the data.

[11] This type of graph has been previously shown to accurately represent the hiring at the firm level (Wu et al. 2018).

[12] The ratios between the standard deviation and the mean for diversity and novelty are 0.242 and 0.255, respectively.

[13] We also test the value-added framework, and the results are similar. We also conduct market value regressions, and the results are directionally similar.

[14] The sample size for columns (1)–(4) in Table 5 is greater than the sample size in Table 4, which includes instrumental variables (IV). When IVs are used in columns (5) and (6), the sample size is the same as in Table 4.

[15] The value-added analysis is similar. Results from market value regressions are also directionally similar.

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